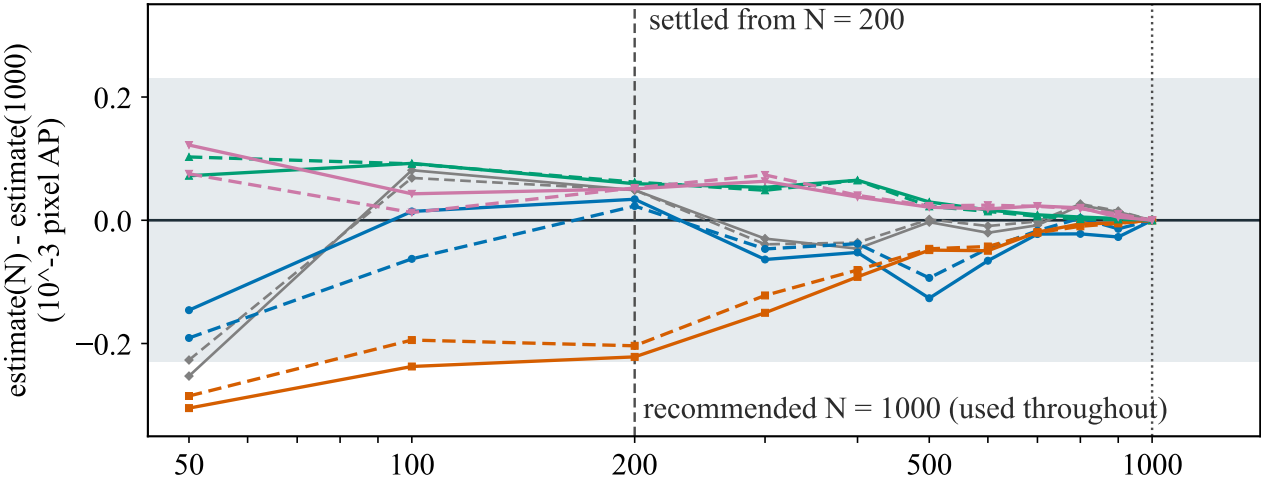


Figure S4. Bootstrap convergence of the interaction: the reported estimates and interval widths have settled.

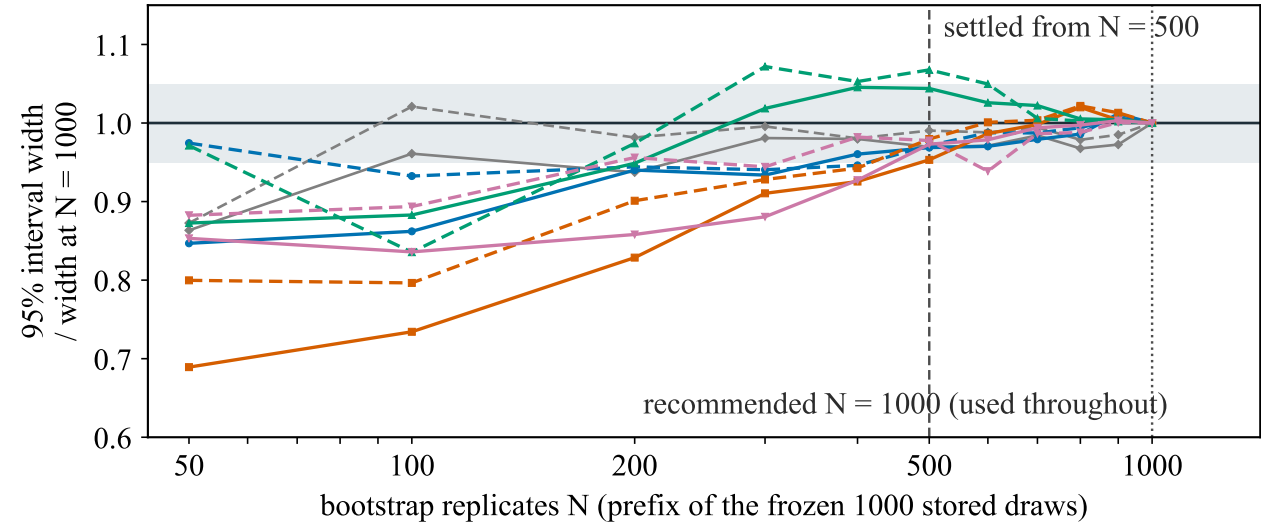
I_TRI and I_BAL are the two interaction contrasts, both in macro pixel AP: $I_TRI = (TRI - DUP)_L - (TRI - DUP)_J$ and $I_BAL = (BAL - A1)_L - (BAL - A1)_J$, with L, J = local, joint matching rule.

—●— MPDD —■— BTAD —▲— MVTec —▼— VisA —◆— KSDD2 (confirmation)

(a) Point-estimate change from its $N = 1000$ value (pixel AP, original metric scale)
solid = I_TRI , dashed = I_BAL ; grey band = $\pm 2.3e-04$ (0 = the $N = 1000$ value)



(b) 95% interval width of pixel AP, relative to its $N = 1000$ value (ratio)
shaded band = $\pm 5\%$; KSDD2 (grey, dashed) = confirmation set, not a family member



No target-domain training, hence no loss-versus-iteration curve to plot.

Prefix bootstrap: the first N of 1000 frozen draws are reused; nothing is resampled.

Estimates: from $N = 200$ within $2.3e-04$ pixel AP; from $N = 500$ within $1.3e-04$.

Interval widths: from $N = 500$ within 6.8% of their $N = 1000$ value (at $N = 200$: 17.1%).

KSDD2 (grey, dashed) is the confirmation set, outside the four-dataset family.

Per- N prefix table and the 10 published cross-checks: figS4_bootstrap_convergence.json.